

Pavan Siva Kumar

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LinkedIn · GitHub · Portfolio

DATABASE ANALYST · SQL · PYTHON · DATA ANALYTICS · DATA VISUALIZATION · CSE '26'

SUMMARY

Computer Science Engineering student with strong foundations in Python, SQL, database systems and data analysis. Experienced building data-driven applications using PostgreSQL, MySQL, MongoDB, Firebase and REST APIs. Skilled in data cleaning, exploratory data analysis (EDA), data visualization, automation and database management using Python, Pandas, NumPy and Matplotlib. Published IEEE research papers with strong analytical and problem-solving skills seeking a Database Analyst role.

EDUCATION	SKILLS
2022–2026 B.Tech — Computer Science Engineering Kalasalingam Academy of Research and Education, Tamil Nadu, Krishnan Kovil CGPA 7.76	LanguagesPython, C, C++, Java, Dart, JavaScript
2020–2022 Class XII — MPC Tirumala Educational Institutions, Bhimavaram 93.7%	AI/MLComputer Vision, CNNs, YOLOv8, Object Detection, Image Classification, Edge AI
2019–2020 10th Standard — SSC Bhashyam School, Bhimavaram 100%	ToolsPyTorch, Ultralytics YOLO, OpenCV, ONNX, Flutter, Jupyter, Google Colab
	IoT Raspberry Pi, ESP32, Arduino, Sensor Integration, IoT Dashboards
	BackendPostgreSQL, MongoDB, MySQL, Firebase, Supabase, REST APIs
	DevGit, GitHub, VS Code, Android Studio, Bridge, Blockbench

PROJECTS
01 Green Guardian GO — ML-Powered Reverse Vending Machine Developed an end-to-end smart recycling system using YOLOv8 on Raspberry Pi for real-time plastic bottle detection and validation. Integrated the ML pipeline with IoT hardware, PostgreSQL backend, and a gamified workflow for session tracking and reward generation. Raspberry PiYOLOv8Computer VisionEdge AIPostgreSQLIoT
02 BluNest — Smart Aquarium Care IoT Application Built a Flutter-based IoT application for aquarium monitoring, sensor data visualization, feeding control, and device status tracking. Integrated Firebase for real-time updates across temperature, turbidity, pH, TDS, and water-level parameters. FlutterFirebaseIoT SensorsML Roadmap3D CAD Design
03 Dual-Mode Robotic Arm Built a robotic arm capable of switching between manual joystick control and AI-assisted pick-and-place operations. Integrated motor control, sensors, and vision-based decision-making for real-time object handling; won 1st Place at the RAS Project Showcase. RoboticsAI VisionMotor ControlSensorsIoT
04 Facial Recognition-Based Smart Attendance System Developed a real-time attendance system using Python, OpenCV, and face_recognition. Automated identity verification and attendance logging with MongoDB integration to reduce manual effort and prevent proxy attendance. PythonOpenCVFace RecognitionMongoDB
05 Minecraft Bedrock Addon Development & Game Scripting Developed custom Minecraft Bedrock addons using JSON-based behavior/resource packs and JavaScript scripting. Designed custom blocks, items, abilities, recipes, textures, and gameplay mechanics while debugging Bedrock API behavior using Bridge and Blockbench. JavaScriptJSONBridgeBlockbenchAPI Debugging

ACHIEVEMENTS	IEEE PUBLICATIONS
01 Finalist, Top 10 of 500+ teams — Ground Reality, Launchpad '25, E-Cell BITS Pilani Hyderabad	01 BluNest: IoT & ML-Based Intelligent Aquarium Management System
02 1st Place — RAS Project Showcase for Dual-Mode Robotic Arm	02 Fix My Ride: ML-Enabled Platform for Intelligent Automotive Assistance
03 Recognized for excellence in competitive engineering events, achieving Top 10 Finalist in two hackathons and securing 6th Place in a Project Expo for innovative software and hardware solutions.	03 Facial Recognition-Based Smart Attendance System Using MongoDB and OpenCV
	04 Dual-Mode Robotic Arm for Manufacturing Industries
	05 IoT Enabled Device-to-Device Communication for Smart City Application